WESTERN GHATS PLANTATION SOIL LABORATORY, CHIKKAMAGALURU

SOIL HEALTH & FERTILITY TEST CERTIFICATE

SAMPLE DATA — FOR PROJECT DEMONSTRATION ONLY (FICTIONAL TEST DATA)

Soil Sample ID:	DEMO-SHC-KA10	**Report Date:**	24-Oct-2026
Location / District:	Mudigere Taluk, Chikkamagaluru District, Karnataka	**Analytical Scheme:**	ICAR-NBSS&LUP; Standard Protocol
Soil Classification:	Humic Acidic Laterite (Plantation Soil)	**Target Agro-Zone:**	Karnataka - Chikkamagaluru Region

SOIL CHEMICAL AND PHYSICAL PARAMETERS

Parameter Name	Symbol	Observed Value	Unit	Reference Range	Rating / Interpretation
Available Nitrogen	**N**	**295.0**	kg/ha	280.0 - 560.0	Medium-High (Rich)
Available Phosphorus	**P**	**22.5**	kg/ha	23.0 - 56.0	Low (Acid Fixation)
Available Potassium	**K**	**235.0**	kg/ha	140.0 - 330.0	Medium (Favorable)
Soil Reaction (pH)	**pH**	**5.10**	pH scale	6.5 - 7.5	Strongly Acidic
Electrical Conductivity	**EC**	**0.12**	dS/m	< 1.00	Normal (Low Salt)
Organic Carbon	**OC**	**1.18**	%	0.50 - 0.75	Very High (Humus Rich)

STANDARD ANALYTICAL READOUT (EXTRACTABLE KEY-VALUE STREAM):

Available Nitrogen (N): 295.0 kg/ha [Medium-High (Rich)]

Available Phosphorus (P): 22.5 kg/ha [Low (Acid Fixation)]

Available Potassium (K): 235.0 kg/ha [Medium (Favorable)]

Soil Reaction (pH): 5.10 [Strongly Acidic]

Electrical Conductivity (EC): 0.12 dS/m [Normal (Low Salt)]

Organic Carbon (OC): 1.18 % [Very High (Humus Rich)]

SOIL HEALTH INTERPRETATION & NUTRIENT ADVISORY

Diagnostic Assessment: High altitude plantation laterite rich in forest humus and organic nitrogen, but strong acidity severely inhibits phosphorus bioavailability and base saturation.

General Nutrient Management Guidance: Apply Dolomite lime @ 2.0 t/ha to correct soil acidity and furnish essential magnesium. Inoculate with Phosphobacteria (PSB @ 5 kg/ha) to mobilize fixed phosphorus. Optimal for coffee, cardamom, arecanut, and black pepper.

Soil Analyst: Demonstration Technical Officer

Quality Authorized by: Senior Agricultural Chemist

Status: Fictional Demonstration Data

ACADEMIC DEMONSTRATION DISCLAIMER: *This soil test report contains fictional demonstration data generated strictly for research, user interface validation, and OCR testing in the MannuMitra AI-Based Predictive Farming System. It is not an official laboratory certificate, does not represent physical soil sampling of any real farming parcel, and must not be used as official government documentation or for commercial agricultural transactions.*